



Round 3 - "Gloves Off"

Welcome to Solvenar! A prosperous and highly developed planet known for technological innovation, a robust economy, and thriving cultural sectors.

This awe-inspiring society will be the stage for the **Great Orbital Ascension Trials** (GOAT). In this Great Galactic Trade-Off, you will face other trading crews head-on as you compete for the coveted title of Trading Champion of the Galaxy. This trading round marks the start of GOAT, where ***all teams begin with zero PnL and the leaderboard is reset.***

You will develop a new Python program and incorporate your strategy for trading **Hydrogel Packs** (`HYDROGEL_PACK`), **Velvetfruit Extract** (`VELVETFRUIT_EXTRACT`), and **10 Velvetfruit Extract Vouchers** (`VELVETFRUIT_EXTRACT_VOUCHER`). These vouchers give you the right to buy Velvetfruit Extract at a later point for a specific strike price.

To kick off GOAT, the Celestial Gardeners' Guild is making a rare appearance, offering you the opportunity to buy **Ornamental Bio-Pods** from them. You may submit two offers and trade with as many of the so-called "Gardeners" as aligns with your strategy for maximum profitability. Secure those Bio-Pods, and they will be automatically converted into profit before the next trading round begins.

Be aware that ***trading rounds on Solvenar (Solvenarian days) last only 48 hours.*** Be decisive, thorough, and fast, and make this first step toward the ultimate title count.

Round Objective

Create a new Python program that algorithmically trades `HYDROGEL_PACK` , `VELVETFRUIT_EXTRACT` , and `VELVETFRUIT_EXTRACT_VOUCHER` on your behalf and generates your first profit in this final phase.

In addition, manually submit two orders to trade Ornamental Bio-Pods with members of the Celestial Gardeners' Guild, then automatically sell your acquired

Bio-Pods to generate additional profit.

Algorithmic trading challenge: “Options Require Decisions”

There are 2 ‘asset classes’ in the three products you trade. The `HYDROGEL_PACK` and `VELVETFRUIT_EXTRACT` are “delta 1” products, similar to the products in the tutorial and rounds 1 and 2. The 10 `VELVETFRUIT_EXTRACT_VOUCHER` products (each with a different strike price) are options, and thus follow different dynamics. All products are traded independently, even though the price of `VELVETFRUIT_EXTRACT_VOUCHER` might be related to that of `VELVETFRUIT_EXTRACT` due to the nature of options.

The vouchers are labeled `VEV_4000`, `VEV_4500`, `VEV_5000`, `VEV_5100`, `VEV_5200`, `VEV_5300`, `VEV_5400`, `VEV_5500`, `VEV_6000`, `VEV_6500`, where VEV stands for **V**elvetfruit **E**xtract **V**oucher, and the number represents the strike price. They all have a 7-day expiration deadline starting from round 1, where each round represents 1 day. Thus, the ‘time till expiry’ (TTE) is 7 days at the start of round 1 (TTE=7d), 6 days at the start of round 2, 5 days at the start of round 3, and so on.

The position limits (see the Position Limits page for extra context and troubleshooting) are:

- `HYDROGEL_PACK` : 200
- `VELVETFRUIT_EXTRACT` : 200
- `VELVETFRUIT_EXTRACT_VOUCHER` : 300 for each of the 10 vouchers.



Example: `VEV_5000` is an option on the underlying VEV with a strike price of 5000 and a position limit of 300. At the start of the final simulation of Round 3, its time to expiry (TTE) is 5 days. In the historical data, the corresponding TTE values are:

- TTE=8d at the start of historical day 0 (coinciding with the tutorial round),
- TTE=7d at the start of historical day 1 (coinciding with Round 1),
- TTE=6d at the start of historical day 2 (coinciding with Round 2).

Vouchers cannot be exercised before their expiry, and inventory does not carry over into the next round. Like in previous rounds, any open positions are automatically liquidated against a hidden fair value at the end of the round.

Manual trading challenge: “The Celestial Gardeners’ Guild”

You trade against a secret number of counterparties that all have a **reserve price** ranging between **670** and **920**. You trade at most once with each counterparty. On the next trading day, you’re able to sell all the product for a fair price, **920**.

The distribution of the bids is **uniformly distributed** at **increments of 5** between **670** and **920** (inclusive on both ends).



Example: counterparties may have reserve prices at 675 and 680, but not at 676, 677, 678, 679, etc..

You may submit **two bids**. If the first bid is **higher** than the reserve price, they trade with you at your first bid. If your second bid is **higher** than the reserve price of a counterparty and **higher** than the mean of second bids of all players you trade at your second bid. If your second bid is **higher** than the reserve price, but **lower** than or **equal** to the mean of second bids of all players, the chance of a

trade rapidly decreases: you will trade at your second bid **but** your PNL is penalised by

$$\left(\frac{920 - \text{avg_b2}}{920 - b2} \right)^3$$

Submit your orders

Submit your two bids directly in the Manual Challenge Overview window and click the "Submit" button. You can re-submit new bids until the end of the trading round. When the round ends, the last submitted bids will be offered to the members of the Celestial Gardeners' Guild.